Srinivasan Poonkundran

(520) 365-5332 | srinivasan.poonkundran@gmail.com | LinkedIn | GitHub | Portfolio

Education

The University of Arizona — Tucson, AZ
Masters in Science, Data Science — *Jan. 2024 – Dec. 2025*

Savitribai Phule Pune University — Pune, India
Bachelor of Engineering, Computer — *Aug. 2017 – May 2021*

Experience

Cognizant Technology Solutions — Sept. 2021 – Jan. 2024
Full Stack Engineer — *Remote*

- Spearheaded performance improvements by optimizing SQL queries and caching, reducing load times by 25%
- Designed and launched a key product feature that improved customer satisfaction by 20%, and revamped UI/UX to boost engagement by 15%.
- Developed scalable microservices using Java and Spring Boot, with backend deployment on AWS and robust database management.
- Created a Python-based PDF label validator, saving $1600 in annual manual effort, and collaborated across Agile teams for iterative development.
- **Skills**: Java, Spring Boot, Unit Testing, Angular, Python, AWS, SQL, Microservices, UI/UX, Performance Optimization, Problem Solving, Attention to detail, Docker, Agile, Cross-functional Collaboration

Projects

Emotion Detection from Text| GitHub

- Built a multi-label BERT-based emotion classifier to detect seven emotions from English text.
- Constructed a custom BiGRU-based Keras model to process frozen BERT embeddings efficiently.
- Applied mixed-precision training and hyperparameter tuning to optimize micro F1-score.
- Created a full inference pipeline using Hugging Face datasets and TensorFlow callbacks.
- **Skills**: NLP, BERT, Hugging Face Transformers, TensorFlow, Keras, Emotion Classification, Multi-label Learning

Heart Rate Monitoring System | GitHub

- Built an IoT-enabled web app to monitor heart rate and blood oxygen levels in real-time.
- Used Node.js, Express, and MongoDB to develop the backend with secure RESTful APIs.
- Created a responsive React frontend to display daily and weekly charts from sensor data.
- Integrated MAX30102 sensor with a Photon board using synchronous state machines.
- Enabled configurable time intervals for data capture and ensured offline storage with sync-on-reconnect.
- **Skills** IoT, Node.js, Express, MongoDB, React, MAX30102, Photon, REST API, Real-Time Charts

Cricket Metrics | GitHub | Website

- Developed a cricket analytics dashboard using R and Shiny to visualize performance from 150,000+ data points.
- Simulated live match data using APIs and integrated the output with a Quarto-based interactive frontend.
- Delivered insights via line charts, bar graphs, donut charts, and map plots for player trends and match metrics.
- **Skills** R, Shiny, Quarto, Data Visualization, Cricket Analytics, API Integration, ggplot2, Dashboard Design

Cricket Match Result Forecasting | GitHub | Website

- Built a Random Forest Classifier to predict ODI match outcomes with 87% accuracy.
- Created a real-time dashboard updating predictions every 5 minutes using simulated match data.
- Developed backend APIs to serve dynamic match stats from historical datasets.
- Improved user engagement with interactive and evolving match outcome predictions.
- **Skills** Random Forest, Python, Dashboard Development, API Design, Cricket Analytics, Real-Time Prediction

Technical Skills

Languages: Python, Java, C/C++, MySQL, MongoDB, JavaScript, Node.js, HTML/CSS, R/R-Shiny
Web Technologies: Angular, React, Spring Boot, JWT, Spring MVC, Maven, RESTful services
Development Practices: Object Oriented Programming, Microservice Architecture, Data Structures
Tools and Framework: R-Studio, Jupyter Notebook, VS Code, Git, GitHub, TensorFlow, Transformers, Docker
Certifications: Amazon Web Services Certified Cloud Practitioner | Credentials